Exponential & Logarithmic Functions Error Analysis

TEACHER KEY Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Log of a sum

ORIGINAL PROBLEM

Evaluate $\log_2(8 + 8)$ without a calculator.

INCORRECT WORKED SOLUTION

$\log_2(8 + 8)$

$= \log_2 8 + \log_2 8$

$= 3 + 3$

$= 6$

A. EXPLAIN THE MISTAKE (1 POINT)

There is no rule that splits the log of a SUM. The product rule, log(ab) = log a + log b, needs a PRODUCT inside the log. Add first: 8 + 8 = 16.

B. SHOW CORRECT WORK (1 POINT)

$\log_2(8 + 8)$

$= \log_2 16$

Since $2^4 = 16$,

$= 4$

2. Log form to exponential form

ORIGINAL PROBLEM

Solve $\log_4 x = 3$.

INCORRECT WORKED SOLUTION

$\log_4 x = 3$

$x = 4 \cdot 3$

$x = 12$

A. EXPLAIN THE MISTAKE (1 POINT)

A logarithm is an exponent. $\log_4 x = 3$ means 4 raised to the 3rd power equals x. The base is raised to the power, not multiplied by it.

B. SHOW CORRECT WORK (1 POINT)

$\log_4 x = 3$

$4^3 = x$

$x = 64$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Fractional base

ORIGINAL PROBLEM

Evaluate $\log_{\frac{1}{3}} 9$ without a calculator.

INCORRECT WORKED SOLUTION

Let $y = \log_{\frac{1}{3}} 9$

$(\frac{1}{3})^y = 9$

$9 = 3^2$

$y = 2$

A. EXPLAIN THE MISTAKE (1 POINT)

The base is $\frac{1}{3}$, not 3. Check: $(\frac{1}{3})^2 = \frac{1}{9}$, not 9. Write $\frac{1}{3} = 3^{-1}$, so $(\frac{1}{3})^y = 3^{-y}$.

B. SHOW CORRECT WORK (1 POINT)

$(\frac{1}{3})^y = 9$

$3^{-y} = 3^2$

$-y = 2$

$y = -2$

4. Power property

ORIGINAL PROBLEM

Write $3 \log_2 x - \log_2 5$ as a single logarithm.

INCORRECT WORKED SOLUTION

$3 \log_2 x - \log_2 5$

$= \log_2 3x - \log_2 5$

$= \log_2 \frac{3x}{5}$

A. EXPLAIN THE MISTAKE (1 POINT)

A coefficient in front of a log becomes an EXPONENT on the argument (power property): $3 \log_2 x = \log_2 x^3$. It is not a factor.

B. SHOW CORRECT WORK (1 POINT)

$3 \log_2 x - \log_2 5$

$= \log_2 x^3 - \log_2 5$

$= \log_2 \frac{x^3}{5}$

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5. Rounding too early

ORIGINAL PROBLEM

An investment grows continuously: $A = Pe^{0.05t}$. How many years until it triples? Round to the nearest hundredth.

INCORRECT WORKED SOLUTION

$3P = Pe^{0.05t}$

$3 = e^{0.05t}$

$\ln 3 = 0.05t$

$1.1 = 0.05t$

$t = 22$

A. EXPLAIN THE MISTAKE (1 POINT)

ln 3 was rounded to 1.1 too early. Keep the full calculator value until the last step, then round: ln 3 ≈ 1.098612, so t ≈ 21.97.

B. SHOW CORRECT WORK (1 POINT)

$3 = e^{0.05t}$

$\ln 3 = 0.05t$

$t = \frac{\ln 3}{0.05}$

$t \approx 21.97$ years

6. Change of base

ORIGINAL PROBLEM

Evaluate $\log_5 40$. Round to the nearest thousandth.

INCORRECT WORKED SOLUTION

$\log_5 40 = \frac{\log 5}{\log 40}$

$\approx \frac{0.699}{1.602}$

≈ 0.436

A. EXPLAIN THE MISTAKE (1 POINT)

In the change-of-base formula the ARGUMENT goes on top and the BASE on the bottom: $\log_b a = \frac{\log a}{\log b}$. Quick check: $5^2 = 25$ and $5^3 = 125$, so $\log_5 40$ must be between 2 and 3.

B. SHOW CORRECT WORK (1 POINT)

$\log_5 40 = \frac{\log 40}{\log 5}$

$\approx \frac{1.602}{0.699}$

≈ 2.292

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7. Same base: distribute the exponent

ORIGINAL PROBLEM

Solve $4^{x+1} = 8^{x-1}$.

INCORRECT WORKED SOLUTION

$(2^2)^{x+1} = (2^3)^{x-1}$

$2^{2x+1} = 2^{3x-3}$

$2x + 1 = 3x - 3$

$x = 4$

A. EXPLAIN THE MISTAKE (1 POINT)

In a power of a power, multiply the exponents: $2(x + 1) = 2x + 2$, not $2x + 1$. Check $x = 4$: $4^5 = 1024$ but $8^3 = 512$, so 4 does not work.

B. SHOW CORRECT WORK (1 POINT)

$(2^2)^{x+1} = (2^3)^{x-1}$

$2^{2x+2} = 2^{3x-3}$

$2x + 2 = 3x - 3$

$x = 5$

8. Isolate the exponential first

ORIGINAL PROBLEM

Solve $2 \cdot 3^x = 50$. Round to the nearest hundredth.

INCORRECT WORKED SOLUTION

$2 \cdot 3^x = 50$

$6^x = 50$

$x = \frac{\log 50}{\log 6}$

$x \approx 2.18$

A. EXPLAIN THE MISTAKE (1 POINT)

In $2 \cdot 3^x$, the 2 is a coefficient, so it cannot be multiplied into the base. Divide both sides by 2 FIRST to isolate 3^x, then take logs.

B. SHOW CORRECT WORK (1 POINT)

$2 \cdot 3^x = 50$

$3^x = 25$

$x = \frac{\log 25}{\log 3}$

$x \approx 2.93$

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9. Extraneous solution

ORIGINAL PROBLEM

Solve $\log_2 x + \log_2(x - 2) = 3$.

INCORRECT WORKED SOLUTION

$\log_2 [x(x - 2)] = 3$

$x(x - 2) = 2^3 = 8$

$x^2 - 2x - 8 = 0$

$(x - 4)(x + 2) = 0$

$x = 4$ or $x = -2$

A. EXPLAIN THE MISTAKE (1 POINT)

$x = -2$ makes the log arguments negative: $\log_2(-2)$ is undefined. Logs accept only positive inputs, so every answer must be checked in the ORIGINAL equation. Only $x = 4$ works.

B. SHOW CORRECT WORK (1 POINT)

$x^2 - 2x - 8 = 0$

$(x - 4)(x + 2) = 0$

$x = -2$: $\log_2(-2)$ is undefined, reject

$x = 4$: $\log_2 4 + \log_2 2 = 2 + 1 = 3$ ✓

Solution: $x = 4$ only

10. Compound interest

ORIGINAL PROBLEM

Find the value of $2,000 invested at 6% compounded quarterly for 5 years.

Use $A = P(1 + \frac{r}{n})^{nt}$. Round to the nearest cent.

INCORRECT WORKED SOLUTION

$A = 2000(1 + \frac{0.06}{4})^5$

$A = 2000(1.015)^5$

$A \approx \$2{,}154.57$

A. EXPLAIN THE MISTAKE (1 POINT)

The exponent is $nt = 4 \cdot 5 = 20$, the total number of compounding periods, not just $t = 5$.

B. SHOW CORRECT WORK (1 POINT)

$A = 2000(1 + \frac{0.06}{4})^{4 \cdot 5}$

$A = 2000(1.015)^{20}$

$A \approx \$2{,}693.71$

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